

CAD-CAE Simulation Analysis Report

Job ID: e80920ce-5093-4823-baef-a01747fe76dd
Project Name: 13073 Combination Wrench v1 I3
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1. Analysis Inputs

Input Parameter	Value / Details
Analysis Types Run	Static Structural, Modal, Fatigue
Material Name(s)	Structural Steel (A36) (E: 200.0 MPa, Nu: 0.26, Density: 7850.0 kg/mm³)
Boundary Conditions	Fixed on Faces: [1] Force on Faces: [2] (fx: 0, fy: 0, fz: -1500.0)
Target Global Mesh Size	2.5 mm
Element Order / Type	Quadratic (Order 2)
Element Count	N/A
Node Count	N/A

2. Simulation Scalar Results

Metric	Value	Unit
Max von Mises Stress	886.60	MPa
Min von Mises Stress	0.00	MPa
Max Displacement	2.9176	mm
Min Safety Factor	0.28	-
Natural Frequencies (Modes 1-3)	815.4 Hz, 5109.9 Hz, 14308.0 Hz	Hz
Min Fatigue Life	1,000,000	cycles

Disclaimer: This report is a decision-support helper. Final engineering decisions require approval by a licensed professional engineer.